

LLMFlux: Batch Processing with LLMs

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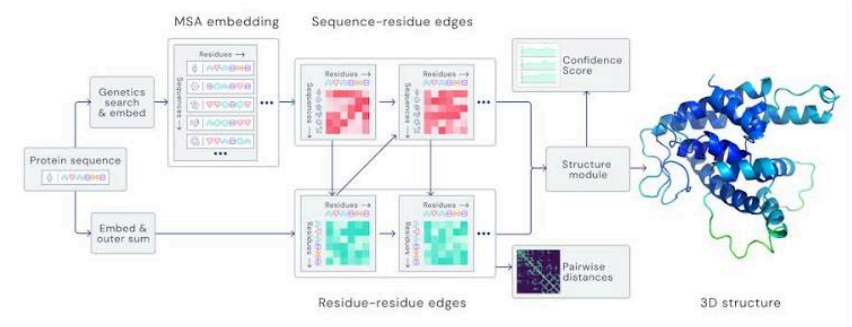
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NCSA AI/HPC Regional Workshop

How AI is used Practically today

- Annotate thousands of photos from the web site
- Protein structure prediction (AlphaFold)
- Generate synthetic data (hundreds of records for each of millions of patients)
- Analyze thousands of imaging files and make diagnostic predictions
- Assist RSEs in writing research software



File after File after File etc.

- Genomics Data
 - Reads data
 - BAMs
 - Vcf
 - BigWig
- Financial Records
- Survey Forms
- Patient Records
- Scan Data
 - CAT
 - MRI



LLM services (at UIUC)

- Illinois Chat
 - Ask a model direct questions
 - Get answers with citations
- LLMHub
 - Always on chat models that you interact with via API or a Frontend
 - Arbitrary models, several model styles
- LLMFlux
 - Runs in the background, applies the same process to many inputs at once, then releases the GPU

Illinois Chat

Docs My Chatbots Create Your Own Bot Sign in

Create a chatbot with your content.

Share it with a click.

Deep search your documents, build an AI-teaching assistant, accelerate your literature review, and get creative.

Try it out →

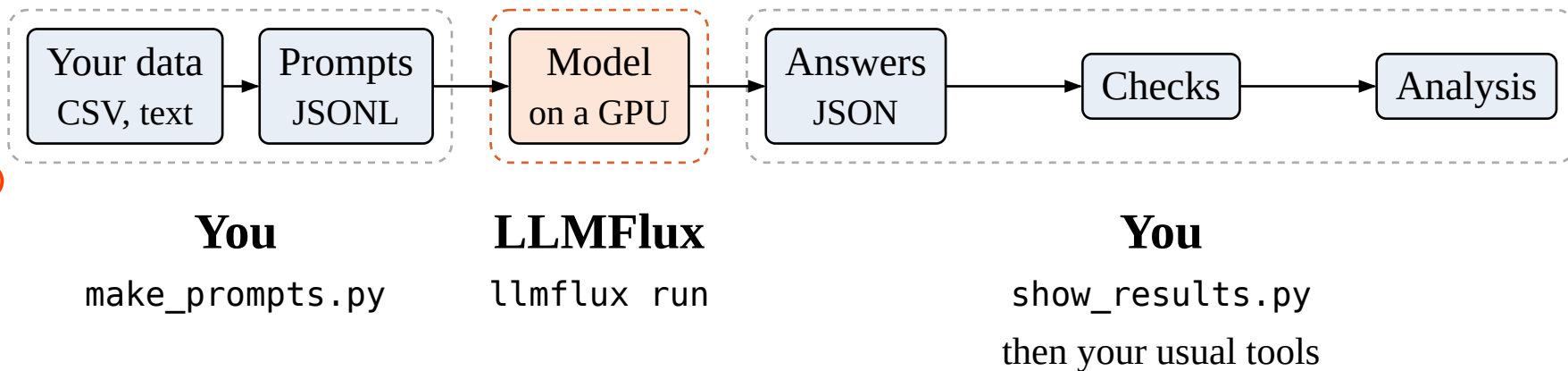
Upload Materials

Drag-and-drop your own files and content

Spring_Syllabus.docx

PDF Word PDF Excel Video Image Audio

Enter Batch Processing



LLMFlux: <https://github.com/Center-for-AI-Innovation/llmflux>

Design Interactively, Run in Batch

- Set up the prompt interactively on a chat for a subset of examples
- Translate the prompt to JSONL
- Process 50,000 samples in one go in bulk

Under the Hood

login node → Slurm → GPU node → results

- SLURM assigns GPU
- Container with the software embedded starts on GPU node
- Inference engine (vLLM) loads the weights in GPU memory
- Each request is processed by the model
- Answers written out in JSON
- GPU node released



Why LLMFlux

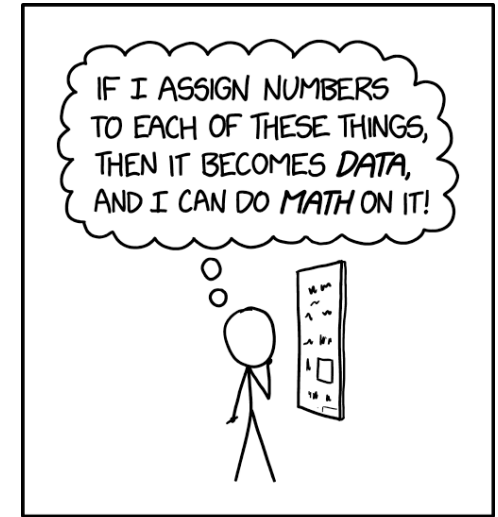
- All the previous can be done manually:
 - Build a Slurm script
 - Find, build, load a container
 - Start an inference engine
 - Query the engine for updates
 - Set up automatic retries
 - Partial progress save on failure
- Or you can run a single command:
 - `llmflux run --model Qwen2.5-7B-Instruct --input prompts.jsonl`

How do I fine-tune the results

- Training on labeled data and fine-tuning is an iterative and expensive process. Try these methods first:
 - Prompt tuning – rewrite the prompt with a couple of examples on the same data, try in a chat window first
 - Give the model full reference text – model input is much cheaper than model output
 - Change model – A larger or more specialized model may work better on your data than a smaller general purpose model
- If those fail, then may want to fine-tune the model:
 - Train a model on your own labeled examples
 - Requires data, separate training methods
 - Pass new Model into LLMFlux via a custom config

Results

- Most of the compute time is loading the model
- Once the model is loaded, it takes only minutes to process many input files
- Batch submission is very efficient



THE SAME BASIC IDEA UNDERLIES
GÖDEL'S INCOMPLETENESS THEOREM
AND ALL BAD DATA SCIENCE.

xkcd by Randall Munroe, CC BY-NC 2.5

Analyzing the Output

- 48/48 replies usable
- Format Checks
 - Checking if it's a valid JSON, requested output appears
 - Can be done programmatically
- Correctness Checks
 - p09 - The method actually contains text from the abstract
 - p06 – Null passed as a string

Try it for yourself!

- Interactive demo: <https://github.com/ncsa/llmflux-workshop>

